Methods & Modeling

Day 4 — Building toward a defined target

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Where we are on the spine

Handoff — participants recap their Day 3 evaluation plan.

- **Yesterday:** data → a credible definition of success.
- **Today:** a definition of success → a model **designed to reach it.**

Day 3: estimand & metrics

Method selection is a choice, not a default

- The method should **follow from** the data (Day 2) and the metrics (Day 3).
- The failure mode to avoid: **reverse-justifying a favored technique** — picking the model first, then arguing for it.
- "I want to use a transformer" is not a method rationale.

The honest order

Most projects run this backward.

The temptation

method → problem → data

The discipline

data → target → **method last**

FIG F4.1 **Method comes last.** A pipeline with the arrow drawn deliberately right-to-left for the "temptation" and left-to-right for the "discipline" — same boxes (data, target, method), opposite order. *See build guide F4.1.*

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THE SPACE OF OPTIONS

The method landscape

The bake-off, not the spectrum

The honest 2026 starting point — and the one that stings:

- On **structured/tabular** clinical data, **gradient-boosted trees still routinely beat deep nets and foundation models** (Grinsztajn et al., 2022).
- Deep nets earn their keep where structure is *inductive-bias-shaped*: sequences, images, signals, irregular multimodal streams.
- "Newer" is a hypothesis, not a default. The spectrum is **capacity vs. inductive bias**, not a ranking.

FIG F4.2 **The bake-off.** A grouped bar chart: penalized regression / GBT / deep net / FM on the same structured-data task — showing the tree ensemble at or above the deep options. (Build from the afternoon notebook.) *See build guide F4.2.*

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Health-specific inductive biases

Health data is **not** generic tabular or text data:

- **Temporality** — events have order and timing that carry meaning.
- **Irregular sampling** — measurements arrive when care happened, not on a grid.
- **Missingness** — *informative*: that a value is missing is a signal.
- **Multimodality** — notes, labs, waveforms, images describe one patient.

FIG F4.3 **What health data demands.** A patient timeline showing irregular, multimodal events with visible gaps — annotated to show order, spacing, missingness, and modality all carrying signal. *See build guide F4.3.*

Worked example — informative missingness

Why "impute the mean and move on" is a modeling decision, not a cleanup step:

- A lab is **missing because it wasn't ordered** — and it wasn't ordered because the clinician wasn't worried.
- The **pattern of measurement** is itself a severity signal.
- Throwing away the missingness mask can discard the most predictive feature you have — or smuggle in leakage.

Day 2: care-process confounding

Day 3: leakage

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THE CENTRAL 2026 DECISION

Foundation models in health AI

Foundation vs. task-specific — the core decision

Foundation model

- Broad pre-training, reused representations.
- Data efficiency downstream; a head start.
- Less control; inherits pre-training biases.

Task-specific model

- Built for one target.
- Wins when the task is narrow, data plentiful, control/interpretability matters.
- No representation to reuse.

Frame it as a **decision**, not an endorsement.

The evaluation crisis for clinical FMs

Grounding, one line: pre-train a reusable patient representation on unlabeled clinical data, then adapt (figure).

The fact this audience should sit with:

- Most clinical FMs are trained on **narrow** corpora and evaluated on tasks that **don't predict health-system value** (Wornow et al., 2023).
- Many **fail to beat strong task-specific baselines** once held to Day 3 rigor.
- "Foundation" names an **aspiration**, not a demonstrated property.

Structured-EHR foundation models are real and improving; the open question is what they're demonstrably *better at*, not whether they exist.

FIG F4.4 **Pre-train then adapt.** Two-stage schematic: large unlabeled clinical data → pre-trained representation → (small labeled task) → adapted model. *See build guide F4.4.*

The real open problems for an EHR FM

Not "is it big" — these are the unsolved, contested questions:

- **Tokenization** of irregular, multimodal, multi-scale event streams — no consensus.
- **Time encoding** — gaps, rates, calendar effects; positional embeddings don't cut it.
- **Multi-site adaptability** — a shared FM that actually transfers across health systems is hard, and recent (Guo et al., 2024).

> Scale buys a representation. It does not buy a contribution — every FM still clears every Day 3 bar.

Day 2: representation is open

Day 3: same bar

Do we still need clinical language models?

The honest counterpoint (Lehman et al., 2023; Agrawal et al., 2022):

- A **general** LLM with good prompting can match specialized clinical models on some tasks — **few-shot clinical extraction** is real.
- ...and a smaller **specialized** model still wins on others (cost, control, latency, privacy).
- The live frontier isn't binary: **PEFT/LoRA** vs. **in-context** sample-efficiency, and **RAG** to ground outputs without retraining.
- The answer is **task-dependent** — which is exactly the point.

Fine-tune, prompt, or train from scratch?

A decision keyed to **data volume, task, and resources:**

- **Train from scratch** — large in-domain data, a task no pre-trained model fits.
- **Fine-tune** — a relevant pre-trained model exists; moderate labeled data.
- **Prompt / in-context** — little labeled data; fast iteration; accept less control.

The rule you *don't* have: **when scale helps is empirical** (measure the data-efficiency curve), and naive **domain-generalization objectives rarely beat plain ERM** (Gulrajani & Lopez-Paz, 2021) — don't assume a fancier objective transfers.

FIG F4.5 **A decision tree.** Branching on: relevant pre-trained model? → labeled data volume? → control/cost needs? → leaf = scratch / fine-tune / prompt. Make it something a participant can trace for their own project. *See build guide F4.5.*

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Representation learning — and where it bites

- A good **representation** is what makes downstream health tasks tractable.
- But representation choices **propagate**: a representation that encodes a shortcut hands that shortcut to every task built on it.
- **Copycats** (Jimenez-Sanchez et al., 2024): a public dataset's quirks get baked in and re-used field-wide.

↩ Day 3: shortcut learning

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BUILDING WELL

Baselines, structure, and trust

Data structure constrains method

- Data **standards** — e.g. **MEDS** — shape what a model can consume.
- Method choice is **entangled** with data representation: you cannot choose a method independently of how the data is structured.
- The data spec you wrote on **Day 2** already narrowed today's method menu.

FIG F4.6 **Representation sets the menu.** A MEDS event stream feeding into the subset of methods that can consume it — the same stream gating which models on the Slide 6 spectrum are even available. *See build guide F4.6.*

Day 2: MEDS

Baselines are the experiment, not the chore

- **Baselines first.** A complex model that can't beat penalized regression hasn't earned its complexity — and this is **routine**, not rare (Wornow et al., 2023).
- A strong baseline is the **null hypothesis** for your method. Reporting only the win is the winner's curse from Day 3.
- Shared, **runnable** baselines (e.g. **MEDS-DEV** on standardized data) are a common currency — so "beats baseline" means the same thing across papers.
- This is **internal validity** — the Day 4 rung of the validity ladder.

Day 3: validity ladder & winner's curse

The afternoon — a fair three-way comparison

The 2:45 notebook, head-to-head on one task:

Logistic regression
the baseline bar

Gradient-boosted trees
the structured-data workhorse

Neural network
capacity, if the data earns it

Same cohort, same metrics (Day 3), same split — so the comparison is honest. The notebook (**Sections 0–7**) then adds a **representation / foundation-model** track and re-checks the **Day 3 shortcut**.

Frontier — uncertainty and abstention by construction

- Build **uncertainty in**, don't bolt it on: calibrated / conformal uncertainty as a *design target*, not a post-hoc patch.
- **Selective prediction** — a model that can *decline* on hard cases beats one forced to answer everywhere.
- This is half of trustworthy-by-construction: failures that **announce themselves** rather than hide.

↩ Day 3: evaluating without ground truth

→ guest: Gerych

Method, data, metric are entangled

- The three cannot be chosen independently — each constrains the others.
- **Trustworthy-by-construction:** build so failures don't hide, rather than auditing them in afterward.
- Tomorrow's question — does it survive a clinic? — starts in how you build today.

→ Day 5: deployment

Propose, then defend

Today's deliverable — Method rationale + 1-slide proposal.
Your **approach**, justified against your **data** (Day 2) and your **metrics** (Day 3) — with a credible **alternative** named.

Next: 11:00 small group — draft your 1-slide method proposal · 1:30 guest (Walter Gerych) — trustworthy-by-construction · 2:45 methods-comparison notebook · 4:15 peer review of proposals.

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